

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I **B Tharun Kumar / 192424356**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	3.0	2.0	1.0	1.0	9.5
Q2	2.5	2.5	2.0	1.0	1.0	9
Q3	2.5	2.5	2.0	1.0	1.0	9
Q4	2.5	2.5	2.0	1.0	1.0	9

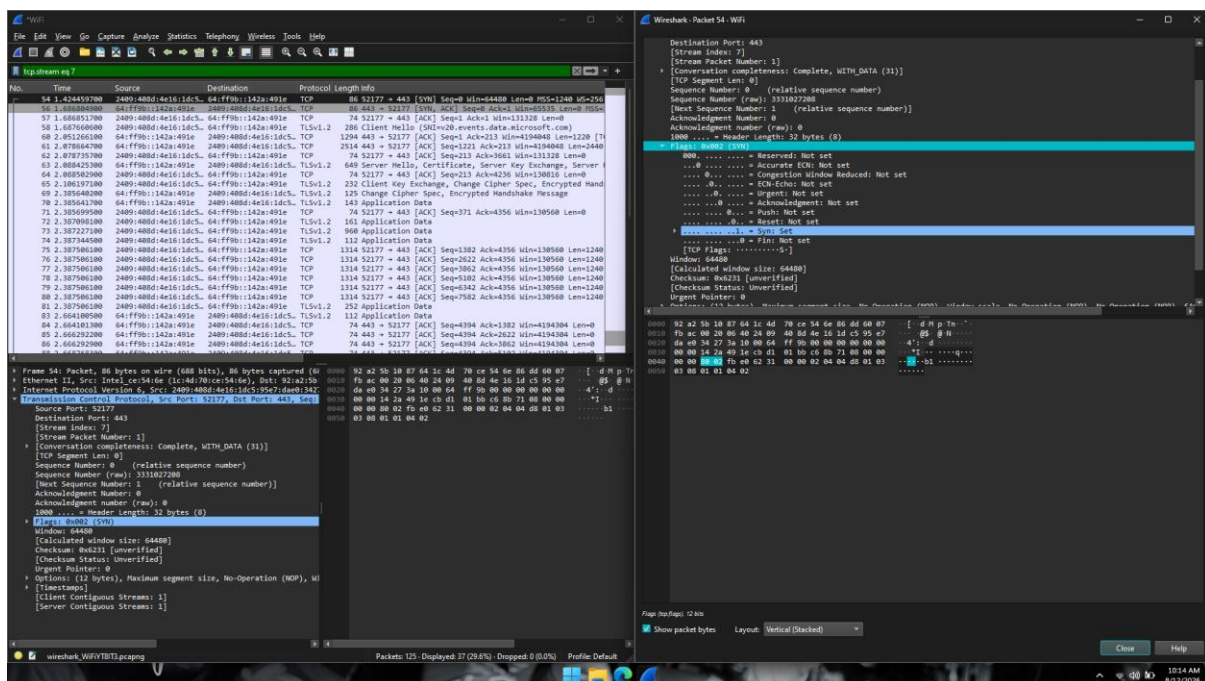
Faculty Signature

Experiments

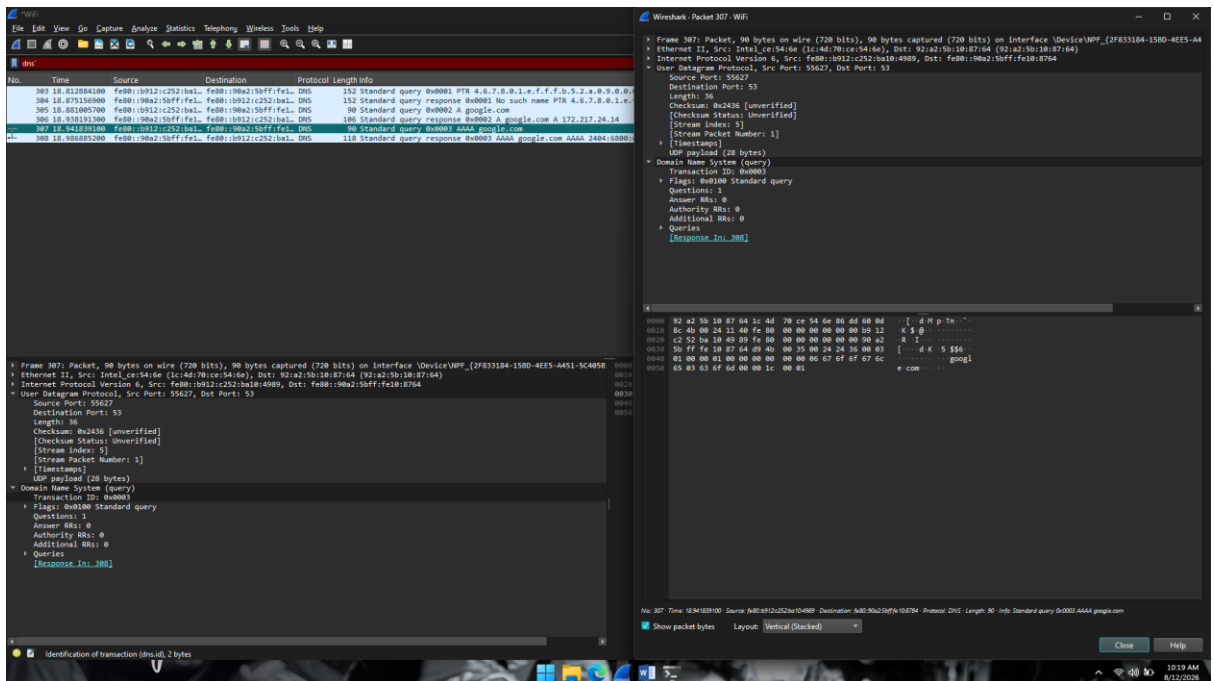
QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4)

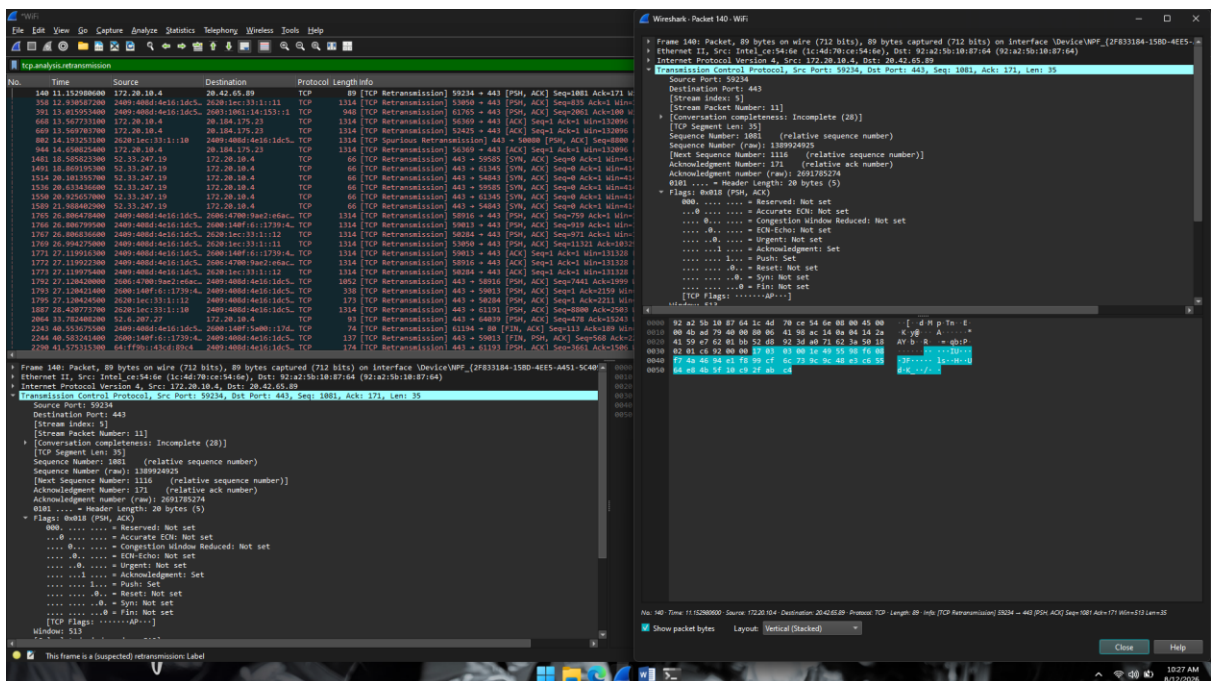


2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability.



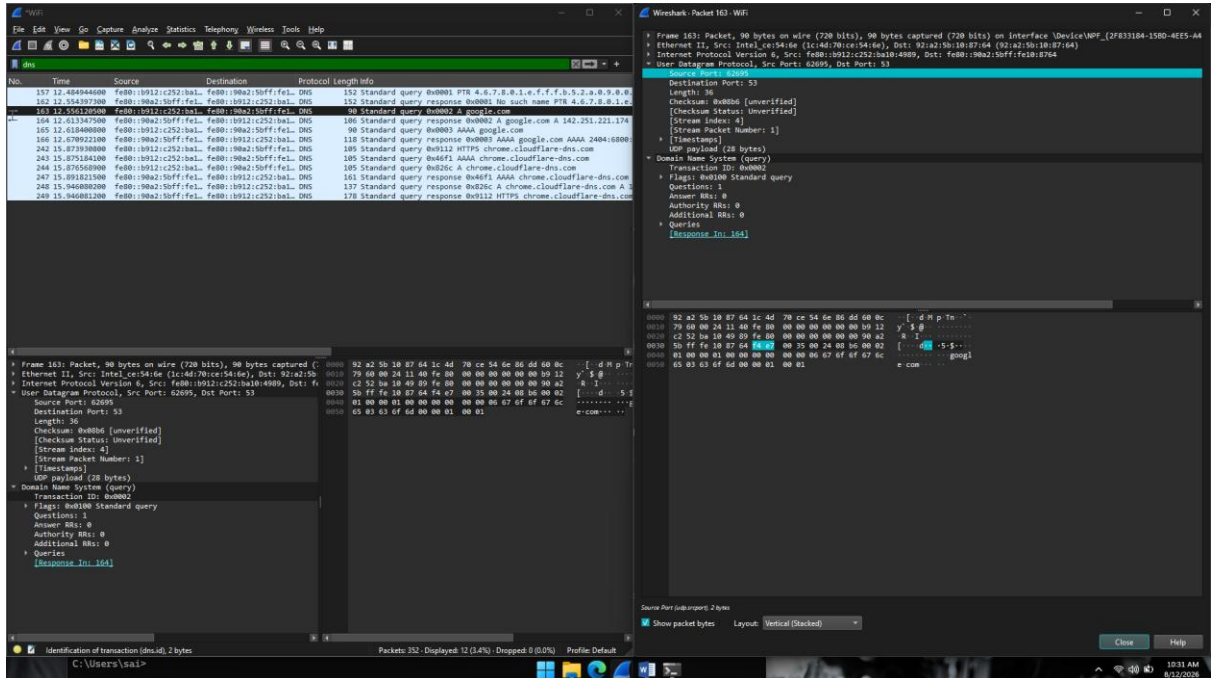
(BL4) (10)

3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP.



The Wireshark capture shows TCP Retransmission packets, indicating that TCP retransmits data when packet loss is detected. TCP uses acknowledgements, sequence numbers and retransmission mechanisms to provide reliable delivery. These mechanisms add overhead and delay, making TCP slower than UDP. UDP does not automatically retransmit lost packets, so it is faster but provides best-effort delivery

4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP



Resolution time

$12.613347500 - 12.561205000 = 0.0521425s$

Packet 163 is the DNS A query for google.com and packet 164 is the corresponding response. The response gives the IPv4 address 142.251.221.174. The DNS resolution time is approximately 52.14 milliseconds. The packet uses UDP with destination port 53. DNS typically uses UDP because it has low overhead, an 8-byte header, and no connection establishment, making DNS communication fast and lightweight

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation